1. **DatabaseMS:** is a collection of interrelated data and a set of programs to access those data. **(DBMS)**

2. **Applications**

- Banking :
- Airlines :
- Universities :
- Credit & transaction :
- Sales :
- Telecommunication :
- Finance :
- Manufacturing :
- Human resources :

3. **Database systems are used to manage that:**

(i) are highly valueable
(ii) " relatively large
(iii) " accessed by multiple users

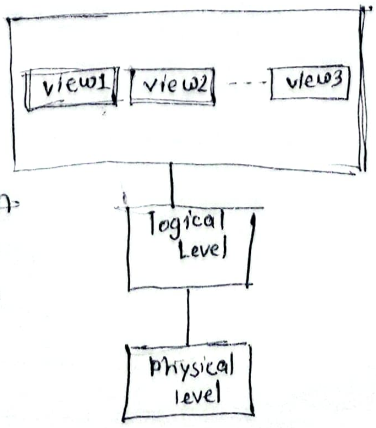
4. **Purpose of Database system.**

5. **- Disadvantages of keeping data in File processing System**

(i) Data redundancy and inconsistency
(ii) Difficulty in accessing data
(iii) Data isolation
(iv) Integrity problems
(v) Atomicity problems
(vi) Concurrent access anomalies
(vii) Security problems

6. **Data Abstraction**

(i) Physical
(ii) Logical



Data abstraction is the process of hiding ~~information~~ unwanted and irrelavant details from the end user.

7. **Instance** : The collection of information stored in the db at a particular moment

Schema : The overall design of the database. is called

8. **Data Models** : collection of conceptual tools for describing data, data relationships, data semantics and consistency constrains.

(i) Relational Model : uses a collection of tables to represent both data
(ii) ER " : Based on the perception of a real world
(iii) Object-based " : combine features of (OBject + Relational) data model
(iv) Semi-structured " : permits the specification of data

9. **DML** is a language that enables users to access or manipulate data as organized by the data model.

Type of access are

(i) Retrieval of information
(ii) Insertion of new info
(iii) Deletion of information
(iv) Modification of "

10. **DDL** : To specify a database schema by a set of definitions

11. **Database Access**

To access the database, DML statement need to be executed from the host language. There are two ways to do this :

- By providing an application program interface that can be used to send DML and DDL to the database and retrieve the result

- By extending the host langueage syntax to embed DML calls within the host langueage program.

Database system module

- Storage manager
- Query processor
- Transaction manager.

Storage Manager

- is the component of database system that provides the interface between low-level data stored in the database and the application programs and queries submitted to the system.

components : (i) Authorization & integrity manager : which tests for
(ii) Transaction manager : which ensures that the dB remains in a consistent
(iii) File manager : which manages
(iv) Buffer manager : which responsible for fetching data from disk storage into main memory.

He implements
- Data files
- Data dictionary
- Indices - can provide fast access to data items.

Transaction Management

Transaction : is a collection of operations that performs a single logical function in a database application.

ACID Properties

Atomicity - A transaction must be fully completed or not executed at all
Consistency - The dB must remain in valid state.
Isolation - Transaction should not interfere with other
Durability - Once a transaction is committed, its changes must persist even in case of system failure.

Example : Transferring funds from account A to B.

DB Architecture

Two-tier Arch : Applications runs on the client machine & interacts directly with dB.
Three " " :
- Client Tier eg web browser
- Application Server
- Database "

⑤ DB users & User Interfaces

- 4 types of users

- Naive users
- Application programmers
- Sophisticated users
- Specialized users

DBA: a person who has such central control over the system is called dB administrator.

Functions of DBA

- Schema definition
- Storage structure and access method definition
- Schema & physical Org. modification
- Granting of authorization for data access
- Routine maintanance